

with any number of internal lines running from one vertex to another, as required to give each vertex the proper number of attached lines. The lines carry arrows as described above, each of which may point upwards or downwards. Each vertex is labelled with an interaction type i and spacetime coordinate x^μ . Each internal or external line is labelled at the end where it runs into a vertex with a field type ℓ (corresponding to the field $\psi_\ell(x)$ or $\psi_\ell^\dagger(x)$ that creates or destroys the particle or antiparticle at that vertex), and each external line where it enters or leaves the diagram is labelled with the quantum numbers $\mathbf{p}, \sigma, n$ or $\mathbf{p}', \sigma', n'$ of the initial or final particle (or antiparticle).

(ii) For each vertex of type i , include a factor $-i$ (from the $(-i)^N$ in Eq. (6.1.1)) and a factor g_i (the coupling constant multiplying the product of fields in $\mathcal{H}_i(x)$). For each line running upwards out of the diagram, include a factor (6.1.9) or (6.1.10), depending on whether the arrow is pointing up or down. For each line running from below into the diagram, include a factor (6.1.11) or (6.1.12), again depending on the arrow direction. For each line running straight through the diagram include a factor (6.1.13). For each internal line connecting two vertices include a factor (6.1.14).

(iii) Integrate the product of all these factors over the coordinates $x_1, x_2, \dots$ of each vertex.

(iv) Add up the results obtained in this way from each Feynman diagram. The complete perturbation series for the S -matrix is obtained by adding up the contributions of each order in each interaction type, up to whatever order our strength permits.

Note that we have not included the factor $1/N!$ from Eq. (6.1.1) in these rules, because the time-ordered product in Eq. (6.1.1) is a sum over the $N!$ permutations of $x_1 x_2 \dots x_N$, each permutation giving the same contribution to the final result. To put this another way, a Feynman diagram with N vertices is one of $N!$ identical diagrams, which differ only in permutations of the labels on the vertices, and this yields a factor of $N!$ which cancels the $1/N!$ in Eq. (6.1.1). (There are exceptions to this rule, discussed below.) For this reason, henceforth we do *not* include more than one of a set of Feynman diagrams that differ only by relabelling the vertices.

In some cases, there are additional combinatoric factors or signs that must be included in the contribution of individual Feynman diagrams:

(v) Suppose that an interaction $\mathcal{H}_i(x)$ contains (among other fields and field adjoints) M factors of the *same* field. Suppose that each of these fields is paired with a field adjoint in a different interaction (different for each one), or in the initial or final state. The first of these field adjoints